

Non-deterministic Model Validation Methodology for Simulation-based Safety Assessment of Automated Vehicles[☆]

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Abstract

Safeguarding and type approval of automated vehicles is a key enabler for their market launch in our complex traffic environment. Scenario-based testing by means of computer simulation is becoming increasingly important to cope with the enormous complexity and effort. However, there is a huge gap when assessing the safety of the virtual vehicle while the real vehicle will drive on the road. Simulation must be accompanied by model validation to ensure its credibility since errors and uncertainties are inherent in every model. Unfortunately, this is rarely addressed in the current literature. In this paper, a modular process is presented covering both model validation and safeguarding. It is characterized by the fact that it quantifies a large number of errors and uncertainties, represents them in the form of an error model and ultimately integrates them into the safeguarding results. It is applied to a type approval regulation for the lane keeping behavior of a vehicle under various scenario conditions. The paper contains a thorough validation of the methodology itself by comparing its results with actual ground truth values. For this comparison, a binary classifier and confusion matrices are used that relate the binary type approval decisions. The classifier demonstrates that the methodology of this paper identifies a systematic error of the simulation model across several safeguarding scenarios. Finally, the paper provides recommendations for alternative configurations of the modular methodology depending on different requirements.

Keywords: Automated vehicles, Model validation, Predictive models, Safety assessment, Systems simulation, Uncertainty aggregation

1. Introduction

In recent years, automated driving has raised great hopes of making future mobility more environmentally friendly, comfortable and safe. Today, vehicles with assistance systems and partial automation are available on the market, but automated vehicles (AVs) from Level 3 onward according to SAE [1] pose great challenges for industry and academia. It is not only challenging to develop AVs, but especially to prove their safety in our complex traffic environment with an infinite amount of different situations [2]. The United Nations Economic Commission for Europe (UNECE) has developed a proposal for the type approval of Automated Lane Keeping Systems (ALKS), which should pave the way for the market release of Level 3 vehicles in the next years [3].

Various safety assessment approaches have been developed by the entire AV research community, since classical real-world testing approaches reach their limits for AVs due to enormously high mileage [4, 5]. Above all, the scenario-based approach is to be mentioned here, as it is frequently used in the literature and as it was implemented in the two large research projects PEGASUS [6] and ENABLE-S3 [7]. It is a promising method that tests the AV in individual traffic situations, mainly by means of systems simulation [8] and by discarding most of the driving time without any actions and events. The scenario-based approach can be accompanied by further safety assessment approaches. System

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functions of the AV can be compared with requirement specifications and standards [9], models of the controller and trajectory planner can be verified by formal methods [10], traffic simulations can analyze the overall impact of AVs on traffic [11], etc. A comprehensive overview of safety assessment approaches can be found in our previous paper [12]. We focus in the following on the scenario-based approach, since it is currently the most widely used approach.

The majority of the safety assessment literature uses mathematical or computational models for their proof of concept, as the real testing effort is hardly feasible. However, the model-based techniques are very rarely accompanied by model validation activities, especially in quantitative terms [12]. This represents a very risky gap. Without validation of the simulation models, there can be no trustworthiness of the models and ultimately no credible decision making for such safety-critical systems as AVs.

In our previous paper [13], we gave a survey about publications across several engineering fields, which present Verification, Validation and Uncertainty Quantification (VV&UQ) approaches to assess the credibility of simulation models. The few papers from the automotive safety assessment field were facing great challenges. They often focus only on individual components such as vehicle dynamics or environmental sensors. They use simple qualitative or deterministic validation approaches without considering different types of errors and uncertainties inherent in any simulation model. They do not analyze the effect of those errors and uncertainties on the final safety assessment decisions. The papers from numerical engineering fields presented enhanced VV&UQ approaches. However, those could, to the best of our knowledge, never be applied to complex practical problems [14] such as cyber-physical systems [15]. Therefore, we introduced a novel, modular and unified VV&UQ framework in [13] and integrated several approaches so that the different engineering fields can benefit from each other.

After presenting the general framework, this paper focuses on its application to the specific use case of safeguarding AVs, on its validation and on the comparison of different framework manifestations. In this paper we are less concerned with the quality of a single simulation model compared to reality and more with the quality of our VV&UQ methodology itself. We therefore apply the Method of Manufacture Universes (MMU) [16], which compares a simulation model with a manufacturing universe to analyze important characteristics. The application of the VV&UQ methodology to the comparison between simulation and reality follows in a subsequent paper.

The main contributions are:

1. a comprehensive overview about the validation of AV models with derivation of requirements,
2. our modular VV&UQ framework in specific configuration for model-based safety assessment of AVs,
3. the first application of a VV&UQ approach with aggregation of errors and uncertainties to AVs,
4. a detailed analysis and fair comparison of a deterministic and a non-deterministic manifestation of the framework based on MMU.

Section 2 summarizes the state of the art in validation of AV models and in VV&UQ in general. Section 3 concludes the state of the art with an extensive analysis and derivation of requirements for our methodology. Section 4 illustrates the use case of the type approval of an automated vehicle in this paper and the configuration of the manufactured universe. Section 5 describes our model-based methodology based on the use case. Section 6 follows a detailed evaluation of the results in order to validate and compare different manifestations of the methodology. Finally, the conclusion in Section 7 summarizes the most important research findings.

2. Related Work

This section begins by outlining fundamental sources of modeling errors and uncertainties to motivate why VV&UQ activities are so important. Then, it gives a comprehensive overview about validation of AV models. Since this field of application is strongly characterized by deterministic systems simulations (point predictions) and hardly any distinction is made between different sources of errors and uncertainties, a non-deterministic approach is presented in the last part. Further information can be found in our survey paper [13]. Here, we highlight individual aspects that are integral to this paper's central theme and understanding².

²Section 2.1 is a compact summary of [13, Sec. 2.2.2, 2.2.4, 2.2.5] and Section 2.3 of [13, Sec. 5.2.3, 6.2.2, 6.4.5, 7.1.1]. Section 2.2 is a restructured and revised version of [13, Sec. 7.2].

Table 1: Classification of references addressing AV model validation. The columns contain the affiliation of the reference to a component according to Sec. 2.2.1-2.2.4. The rows distinguish between deterministic and non-deterministic simulation. References of the same research group are combined in the same bracket.

	Sensor	Vehicle	Closed-loop	Traffic
Deterministic	[22], [23, 24], [25], [26], [27]	[28, 29], [30]	[31, 32, 33], [34], [35]	[36], [37], [38], [39]
Non-deterministic	-	[40, 41], [42]	[43, 10], [44]	[45]

2.1. Sources of Errors and Uncertainties

Errors are inherent in every simulation model, since a model is by definition a simplified abstraction of reality. The error, denoted e , indicates the deviation compared to the true value of nature y_{true} . However, it is often quite difficult to precisely quantify the errors of a simulation model. As soon as it is not possible, associated uncertainties arise. Two basic types of uncertainties can be distinguished. Aleatory uncertainties describe stochastic effects and natural variability and can be quantified by probability distributions. Epistemic uncertainties exist when knowledge is poor and can therefore either be reduced or at least be quantified, e. g. by intervals treating all values equally. [17, Sec. 1.2].

One source of errors and uncertainties results from the calculation of computational models with computers of finite precision compared to the exact solution of the mathematical model. These numerical errors can vary greatly depending on the specific application. They include both basic code errors and errors in solving the equations, such as rounding errors or the discretization error e_h . During model verification, the first category is treated by code verification and the second category by solution verification [18, Sec. 2].

Another source of errors and uncertainties arises from model inputs (e_x) and parameters (e_θ). Whereas they are usually assumed fully characterized in deterministic simulations, this is not the case for partially and uncharacterized experiments as well as unknown model prediction conditions. Input uncertainty quantification intends to rigorously quantify those uncertainties. In non-deterministic simulations, the input uncertainties are propagated through the simulation model and aggregated on the output side [19].

Furthermore, the model-form itself contains errors e_m and uncertainties due to assumptions or the selection of an inadequate equation. They can be quantified based on physical data from model validation experiments. The experiments are re-simulated and the results of the physical system g_s and the simulation model g_m are compared using validation metrics [20].

The involved errors can be summarized as follows [18, eq. (1-5-6)]:

$$y_{true} = g_s(x) - e_{y,obs} \quad (1)$$

$$= g_m(x, \theta, h) - (e_m + e_x + e_\theta + e_h). \quad (2)$$

The various sources of errors and uncertainties are deeply interwoven and interfere with each other. Depending on the constellation, they can magnify or compensate each other, which results in a misleading trustworthiness of the simulation model [18, Sec. 1]. This effect is sometimes seen during model calibration, when an inadequate model-form is compensated by adjusting the parameters far beyond their physical significance until the results coincide under certain calibration conditions [21]. It is therefore helpful to quantify the different uncertainties separately as far as possible. Certainly, additional sources such as measurement errors $e_{y,obs}$ or extrapolation errors caused by model predictions outside the scope of validity make this even more difficult.

2.2. Validation of Automated Vehicle Models

We sort this section on the validation of AV models based on different components that form the entire AV, since there exists some literature on component-level, but just a few references on system-level. The central modules of an AV include environmental sensors, several software units for environment perception, motion planning and control, as well as actuators and vehicle dynamics. An overview about most of the references is given in Table 1.

2.2.1. Sensor Model Validation

The development of models for camera, lidar and radar sensors is currently a key enabler for simulation-based safety assessment of AVs. They are essential for realistic simulations and were recently used to identify challenging test

scenarios for AVs [46]. Holder et al. [47], Rosenberger et al. [48] systematically derive requirements for sensor models to determine required fidelity levels. Schaermann et al. [24] distinguish parametric, non-parametric and ray-tracing approaches for sensor modeling, either on the level of raw sensor data or of processed object lists.

Regarding sensor model validation, it is also possible to distinguish between validation on raw data or object list level. Hanke et al. [23], Schaermann et al. [24] compare a lidar sensor and its corresponding model with regards to raw point clouds and occupancy grids of traffic objects. For the comparison, they select three validation metrics that can handle the complex nature of the data. They are referred to as overall error, Barons and Pearson correlation coefficients. Abbas et al. [22] develop a metric to compare the visual complexity of real and synthetic images as camera raw data based on color and spatial information.

Nentwig et al. [26] apply a classifier to camera images to perform comparisons on the level of object lists. They compare both the object hypothesis and the bounding boxes from the classifier, the latter based on the box sizes and the former based on a confusion matrix. Gaidon et al. [25] also apply classifiers and present eight metrics to measure the real-to-virtual performance gap. Zec et al. [27] validate the sensor fusion of camera and radar models. On the one hand, they compare time signals of processed traffic objects with a log-likelihood and a Root Mean Square Error (RMSE). On the other hand, they compare statistical histogram data with a Jensen-Shannon divergence.

2.2.2. Vehicle Model Validation

For validation of vehicle dynamics models, standardized test maneuvers such as steady-state cornering [29] or sine with dwell are used [28]. They provoke a characteristic behavior of the vehicle, which can be evaluated on the basis of time signals and Key Performance Indicators (KPIs). Both standards were developed with regard to the simulation-based type approval of electronic stability control systems. The standard [29] assumes one deterministic simulation of the steady-state cornering maneuver as a baseline, which results in one time signal of the steering wheel angle, the sideslip angle and the roll angle, plotted against the lateral acceleration, respectively. It adds a pre-defined tolerance band around each baseline signal and checks whether multiple repetitions of physical experiments fall within these bands. Afterwards, the standard [28] performs a similar comparison based on tolerances for specific KPIs in the sine with dwell maneuver. If all tolerances are met, the generic validity of the vehicle dynamics model is concluded. Then, the actual type approval of different vehicle variants can take place in the simulation [49]. Kutluay and Winner [30] give a comprehensive overview of the literature in validation of vehicle dynamics models.

Kutluay [40] inverts the tolerance approach from those standards in his PhD thesis. Instead of adding the tolerances to the deterministic simulation, he calculates confidence intervals from the experimental repetitions based on Student's t-distribution and adds the tolerances to the confidence intervals around the experimental mean. He offers either absolute magnitudes or relative percentages for the tolerance values. Then, he can check, whether the simulation falls within the area around the experimental mean. He distinguishes an averaged input case, where the experimental input data is averaged to perform one deterministic re-simulation, and an averaged output case, where each experiment is re-simulated and the average is taken on the result stage. According to Oberkamp and Roy [50, p. 492], the former will lead to significant deviations for strongly non-linear systems.

Viehof [41] extends the validation methodology from Kutluay [40] in his PhD thesis. He also re-simulates each experimental repetition, but then compares Probability Density Functions (PDFs), without averaging the outputs. He accepts the simulation model based on a statistical t-test, if the PDF from the model lies within the one from the experiment. In case of lower requirements, he switches back to the conventional tolerance values and checks whether the simulation PDF lies within the tolerance around the experimental mean. Similarly, Rhode [42] use non-deterministic Monte Carlo simulations and confidence bands for vehicle dynamics model validation.

2.2.3. Closed-loop Model Validation

In our previous paper [35], we used an adaptive cruise control function to control a test vehicle and its corresponding vehicle model. We compared a purely virtual Model-in-the-Loop (MiL) simulation and a hybrid Vehicle-in-the-Loop (ViL) simulation on a chassis dynamometer with physical proving ground tests. As test scenarios, we selected vehicle following, emergency braking and cutting-in. We used qualitative graphical comparisons and quantitative validation metrics such as the correlation coefficient to validate the MiL and ViL simulation along the control pipeline.

Groh et al. [31], Notz et al. [32], Wagner et al. [33] focus on the influence of the sensor data on the validation of the closed-loop AV, compared to independent ground truth measurements of the environment. They compare signals such

as the velocity, trajectories or an overall risk measure along the control pipeline and evaluate the validation errors via box-plots. Aramrattana et al. [51] analyze the influence of modeling errors and uncertainties on control performance.

Schürmann et al. [10] use formal methods for safety assessment of AVs. They create non-deterministic models based on set theory and apply reachability analysis to determine which states the AV can reach from given initial states and possible inputs and parameters. If the reachable set of the AV does not intersect with predicted ones of other traffic participants, the AV is safe. In contrast to sampling techniques such as Monte Carlo, the reachability analysis can formally guarantee safety. They require non-deterministic models for the vehicle dynamics with controller and for the other traffic participants.

Hartung et al. [43] create those models by using a conformance testing technique. They define a formal notion of conformance, similar to validation metrics, and test whether the behavior of the non-deterministic model encloses the one from the physical system. Strictly speaking, the conformance testing is formulated as an inverse optimization problem, which optimizes the parameter sets so that the bound is as tight as possible. This can be seen as a model calibration approach rather than a model validation approach. Böde et al. [44] propose three notions of model validity and find an optimal split between calibration and validation data. Johnson et al. [34] apply formal methods based on correct-by-construction controller design and validate the closed-loop model by comparing the percentage of collision-free runs in a parking lot scenario.

2.2.4. Traffic Model Validation

In contrast to microscopic systems simulations, traffic simulations with multiple agents do not assess individual AVs, but determine the overall impact of AVs on traffic. Detering et al. [45] measure parametric uncertainties for Monte Carlo simulations. Zheng et al. [52] use extreme value theory to validate statistical traffic models. Rao and Owen [38] apply autoregressive integrated moving average models to analyze the errors of traffic models. An overview about the calibration of traffic models including metrics for comparison is given in [36, 37, 39].

2.3. Probability Bound Analysis

Oberkamp and Roy [50] use Probability Bound Analysis (PBA) as VV&UQ framework to separately quantify input, numerical and model-form uncertainties. They describe aleatory uncertainties in the form of probability distributions, epistemic uncertainties as intervals and mixed uncertainties as probability boxes (p-boxes). The latter are imprecise probabilities that add an interval width to a Cumulative Distribution Function (CDF). By combining all sources of uncertainties they obtain a final p-box, which bounds the true value with a high probability.

They use the Richardson extrapolation technique to quantify the discretization error by comparing results of different step sizes as a replacement for the exact mathematical model. They quantify input uncertainties and apply nested Monte Carlo sampling to propagate them through the simulation model. In the outer loop they take samples from the epistemic parameters and in the inner loop from the aleatory parameters for each of the epistemic ones. While all aleatory samples form a single CDF – more precisely a stepwise Empirical CDF (ECDF) – the epistemic samples provide the width of the CDF and ultimately result in a p-box. During model validation, they quantify the area between the distribution from simulation and experiment to quantify the model-form uncertainty. They extrapolate the latter from the experimental validation conditions to the ones used for model prediction with a regression model. Finally, they add the epistemic intervals from the model-form and the numerical uncertainty to both sides of the input uncertainty distribution to obtain the final p-box.

3. Analysis of the Related Work

In this section, we analyze the related work presented so far to identify research gaps and directions for the remaining paper. Again, we address both the validation of AV models and the PBA approach.

3.1. Validation of Automated Vehicle Models

The related work on the validation of AV models contains references addressing specific component models as well as references addressing the entire closed-loop model:

1. The sensor model references focus mainly on the complex nature of the multi-dimensional sensor data and on specific sensor effects. Comparisons on the level of object lists contain the impact of the sensor data on the detection algorithms and allow the application of typical time series metrics.
2. The vehicle dynamics references are based on standardized maneuvers and tolerance values for the permissible deviations between simulation and experiment. These tolerances represent subjective expert estimates with regard to a classic vehicle dynamics evaluation and describe model validity as a simple binary result. Regarding AV safety, these tolerances are not applicable, since model validation by definition always refers to a certain use case.
3. The current closed-loop references are either embedded in a formal framework running online in the AV or analyze how modeling errors flow through the AV pipeline.
4. The traffic model references are interesting for safety assessment approaches that use traffic simulations to analyze the impact of AVs. Since we are focusing on the scenario-based approach, we require a re-simulation of traffic trajectories instead of traffic models.

All approaches lack an aggregation of errors and uncertainties for the safety assessment of the entire AV. It is crucial to analyze the impact of these errors on safety, because in the end we are interested in making a decision on the safety of the entire AV based on the simulation models.

The validation of component models is a powerful tool for the developer of the respective component. Furthermore, it also increases the trustworthiness of the entire model on a qualitative level. On a quantitative level, however, it is very difficult to relate the errors in component modeling to the final decisions about the system. Suppose a robust vs. a sensitive lane keeping function controls the same vehicle. The robust controller might compensate a poor vehicle dynamics model violating the permissible tolerances. The sensor, controller and vehicle dynamics interact with each other and should not only be validated independently.

This type of extrapolation in the system hierarchy from component to system level is generally one of the main current challenges in VV&UQ [13, Sec. 6.6] with only a few recent approaches such as [20, 17]. There are engineering fields such as spacecrafts [53] or nuclear reactor safety [54] that make system level tests impossible for cost or safety reasons and require such VV&UQ approaches. However, this is not the case with AVs. Therefore, we concentrate in this paper exclusively on the entire AV. Nevertheless, we make sure that the methodology is also applicable to component models.

3.2. Probability Bound Analysis

In our survey paper [13], we have also presented a variety of references that focus on advanced non-deterministic approaches. However, there were only a few general approaches which cover the entire model-based process and which proved difficult to apply to complex systems. For example, Eek et al. [15] found that the number of components and parameters of an aircraft simulation model is too complex to be handled by PBA with reasonable effort. Nevertheless, we presented a modular framework in [13] and are firmly convinced that the advanced approaches can be used in selected blocks of the framework even for complex systems such as AVs.

In this paper, we have chosen to use PBA as part of our framework for safety assessment of AVs. PBA is the main approach of Frequentist VV&UQ. It objectively estimates relative frequencies of model inputs and parameters based on repeated measurements. Unlike Bayesian approaches, it does not incorporate personal beliefs in the form of prior probabilities, nor does it modify the original model with new data based on Bayes' theorem. Therefore, it meets the requirements of an independent type approval of an automated vehicle, which we are particularly interested in.

4. Use Case

In the first part of this section, we present the Lane Keeping Functional Test (LKFT) of UN regulation 79 for the type approval of lateral driving functions. In the second part, we describe the selection of our simulation models based on the Method of Manufactured Universes.

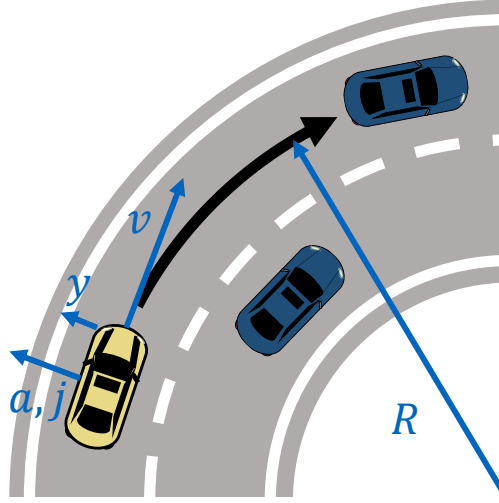


Figure 1: Lane Keeping Functional Test (LKFT)

4.1. Lane Keeping Functional Test

We choose UNECE regulation 79 in Revision 4 [55] as the use case of this paper and target the Automatically Commanded Steering Function (ACSF) of category B1. The latter "means a function which assists the driver in keeping the vehicle within the chosen lane, by influencing the lateral movement of the vehicle." In production vehicles, it is either a standalone assistance system (SAE Level 1) or combined with adaptive cruise control for Partial Automation (SAE Level 2). The R-79 lends itself as a use case, because a regulation generally has a public and independent character with clearly defined test cases and pass/fail criteria, and the R-79 is also binding for current production vehicles of Level 1 and 2.

The R-79 describes the Lane Keeping Functional Test (LKFT) [55, Sec. 3.2.1] to check whether the ACSF system can detect and keep its own lane without running over the lane markings. The principle is visualized in Fig. 1. In detail, the test scenario is specified as follows:

- "The vehicle shall be driven [...] with a constant speed on a curved track with lane markings at each side."
- "The necessary lateral acceleration to follow the curve shall be between 80 and 90 per cent of the maximum lateral acceleration specified by the vehicle manufacturer $a_{y,max}$."
- "The vehicle speed shall remain in the range from v_{min} up to v_{max} ."
- "The vehicle manufacturer shall demonstrate [...] that the requirements for the whole lateral acceleration and speed range are fulfilled."

The pass criteria are fulfilled if:

- "The vehicle does not cross any lane marking;"
- "The moving average over half a second of the lateral jerk does not exceed 5 m/s^3 ."

Thus, the LKFT focuses on the stationary conditions of a roughly constant lateral acceleration a and velocity v . The ranges depend on the manufacturer and are assumed here as typical values $v_{min} = 80 \text{ km/h}$, $v_{max} = 180 \text{ km/h}$ and $a_{y,max} = 2.5 \text{ m/s}^2$. The curve radius $R = v^2/a$ is a third dependent variable. We extend the LKFT in this paper by the scenario parameters wind speed v_w , road gradient s_r and tank load l_t . We thus bring the example closer to reality and consider parameters from different categories without losing focus on v and a . The categories include driving states of the vehicle as well as the road and the environment layer of the 5-layer environmental model presented in [56]. Our LKFT example can be summarized in vector notation as

$$\mathbf{x} = \begin{bmatrix} 1 & v & a & v_w & s_r & l_t \end{bmatrix}^T \in \mathbb{R}^{N_x+1} \quad (3)$$

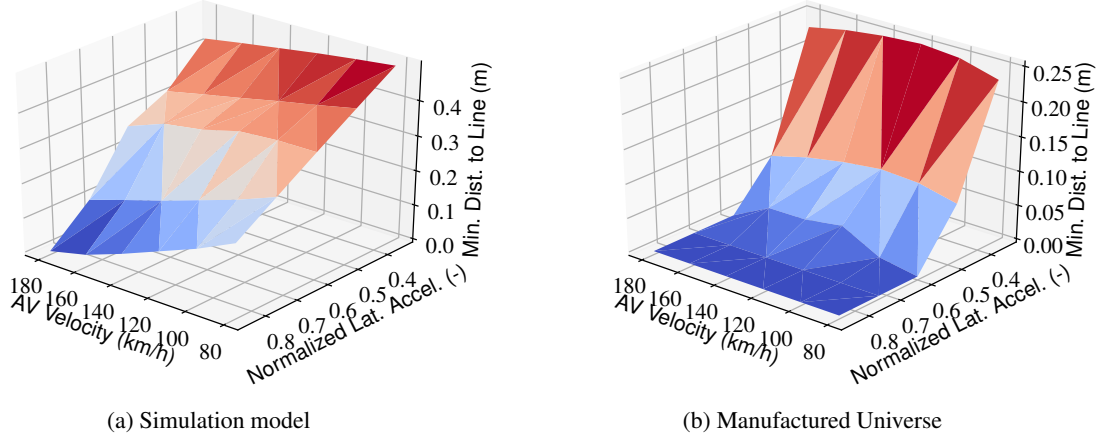


Figure 2: Minimum distance to line across the scenario space

with $N_x = 5$ inputs. We concentrate on the $N_y = 1$ output of the distance to line $y \in \mathbb{R}$, as it is much more descriptive than the jerk. Finally, we are aware that the regulation with two parameters is currently pursued with physical testing. Nevertheless, the extended use case serves as a blueprint for a model-based safeguarding process, as it is firmly planned for higher automation levels [3, 6]. Therefore, we will use the term AV as a placeholder for all automation levels.

4.2. Method of Manufactured Universes

Stripling et al. [16] introduce the Method of Manufactured Universes (MMU) as an approach to validate uncertainty quantification methods. Thus, it is not a method to validate the models, but to validate the validation methodology itself. The intention of MMU is to create a manufactured universe where the true values of nature are known. In contrast to reality, where the true values can be estimated, if at all, by extensive measurements, the manufactured universe impresses by any number of simulations in a known environment. For sure, the user must manufacture the universe in a way that it is close to reality. Otherwise, the transferability of the findings is not guaranteed. In MMU, the actual simulation model is compared with the manufactured one by means of the desired validation methodology. Stripling et al. [16] analyze UQ methods in a particle transport universe by comparing the actual low-fidelity model with a high-fidelity reference. Whiting et al. [57] create a CFD universe to compare four validation methodologies including PBA under validation and prediction conditions.

In accordance with the current state of science, we also create a universe for the validation of our methodology. We adapt the MMU approach by injecting modeling errors into the manufactured universe so that we can verify that a validation methodology is able to identify these errors. We select a pre-configured vehicle dynamics model of a sports car, a simple PI controller for lane keeping and an ideal sensor model from a vehicle simulation tool. We use a vehicle mass of 1377 kg for the simulation model and a mass of 1577 kg for the manufactured universe (MU) model. This systematic error of 200 kg can be compared with the weight of two to three passengers. The mass has an influence on the overall driving behavior of the vehicle. The resulting minimum distance to line across the scenario space is already anticipated in the two surface plots in Figure 2. Higher lateral accelerations and higher velocities for constant radii lead to a smaller distance to line. This corresponds to the driving physics during cornering in which the vehicle is pushed outwards by the centripetal force. The heavier vehicle has smaller distances and for high lateral accelerations it has even a plateau of zero distance to line representing a line crossing. We deliberately generated this behavior in our universe because it is especially safety-critical if the simulation model passes all tests and gives the developer a positive feedback although the real system would fail several times. It is now the challenge for the validation methodology to identify those fails.

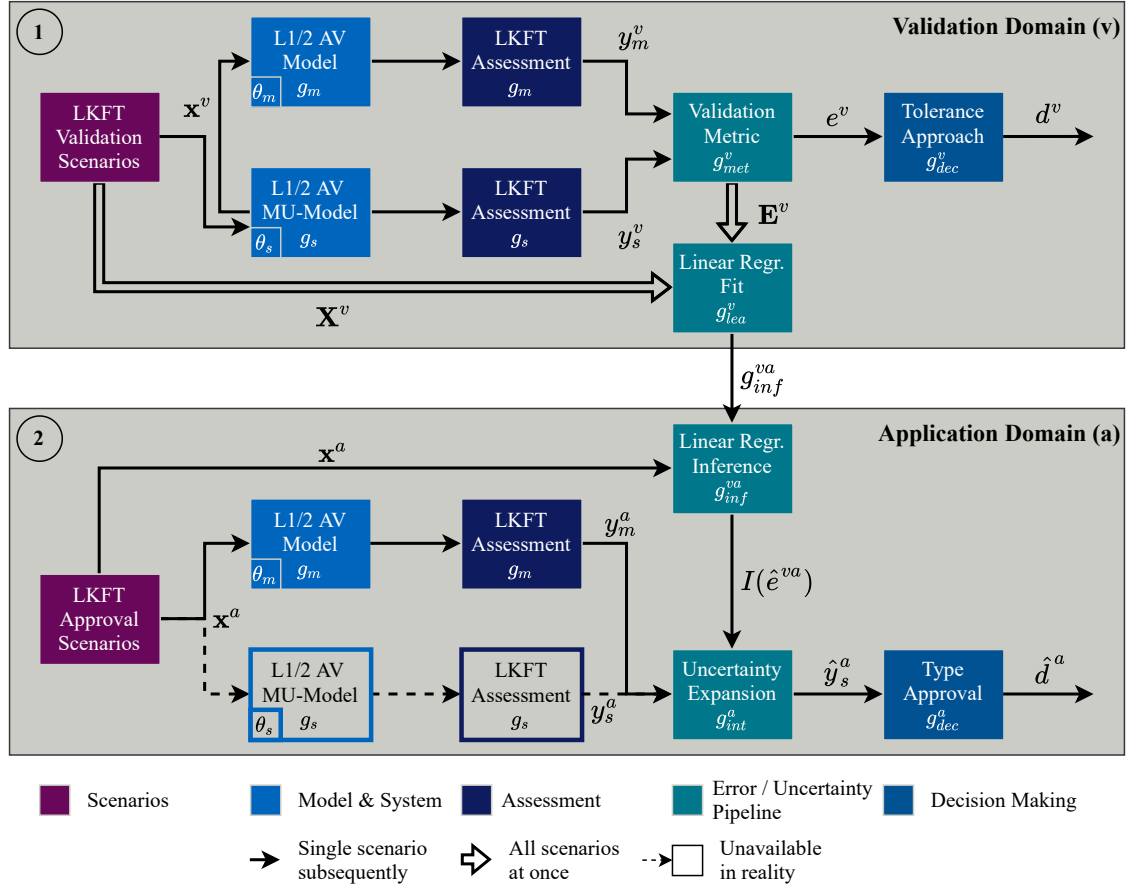


Figure 3: Model-based process for AV homologation, based on previous work in [13, Fig. 1]

5. Methodology and Application Results

In this section we implement our generic framework from [13, Fig. 1] for the use case of safeguarding AVs. The specific configuration of the framework can be seen in Fig. 3. It describes a model-based process of model validation and model prediction in the application domain with several individual steps. We describe the methodology step-by-step from left to right and from top to bottom, using exemplary results of the LKFT from R-79. The mathematical symbols that describe the interface of each framework block are summarized in Fig. 3. We deal with both a deterministic and a non-deterministic manifestation of the framework during the description of the blocks and will compare them in the next section. The two terms refer to the type of simulation. In contrast to state of the art deterministic validation, our deterministic manifestation also takes interpolation uncertainties into account, but no input uncertainties compared to the non-deterministic simulation. The interested reader is referred to [13] regarding general information and more detailed theory.

5.1. Model Verification

Before proceeding with model validation and prediction, the numerical effects during model verification should be analyzed and the model parameters estimated. We skip the step of model calibration, since we assume already parameterized models for an independent type approval. Therefore, we dedicate this subsection to a numerical pre-analysis.

Since we do not have the exact mathematical model, we use the Richardson Extrapolation technique to estimate the numerical discretization error. We perform three simulations with a fine step size $h_1 = 2.5 \times 10^{-4}$ s, a medium step size $h_2 = 5 \times 10^{-4}$ s and the actual coarse step size $h_3 = 1 \times 10^{-3}$ s. This corresponds to a small refinement factor

Table 2: Parameter characteristics. There are several scenarios in the global validation and application space and again several random samples in the local space around each scenario. A normal distribution $\mathcal{N}(\mu, \sigma^2)$ is specified by its mean μ and variance σ^2 . The repetitive samples refer to the physical validation experiments, the nested samples to the remaining non-deterministic simulations.

Parameter	Unit	Validation			Application			Uncertainty (around each scenario)			
		space		scen-arios	space		scen-arios	space		samples (nested)	samples (rep.)
		min	max		min	max		type	size		
Velocity	km/h	90	170	3	80	180	6	Aleatory	$\mathcal{N}(0, 0.5)$	$\Sigma 10$	$\Sigma 10$
Lat. Accel.	-	0.4	0.8	3	0.35	0.85	5	Aleatory	$\mathcal{N}(0, 0.01)$		
Wind Speed	km/h	-5	5	2	-5	5	2	Aleatory	$\mathcal{N}(0, 2)$		
Tank Load	kg	-20	20	2	-20	20	2	Aleatory	$\mathcal{N}(0, 0.5)$		
Road Slope	°	-1	1	2	-1	1	2	Epistemic	$[-0.1, 0.1]$	3	
Π		$N^v = 72$			$N^a = 240$					30	$N_r^v = 10$

$r = h_3/h_2 = h_2/h_1 = 2$. It gives identical results of the distance to line in the relevant decimal places for all three step sizes. This shows high convergence and very small numerical errors $e^n \approx 0$.

The negligible numerical effects coincide with the findings of Viehof and Winner [58] from vehicle dynamics simulation. However, we would like to point out that complex co-simulations with different solvers and step sizes are sometimes used to assess the safety of AVs, which no longer necessarily lead to similar results. Therefore, this should be investigated at least once per tool chain.

5.2. Scenario Design

Since the LKFT requires an assessment over "the whole lateral acceleration and speed range", we concentrate directly on a good coverage of the scenario space, without restricting us only to the critical band from 80 to 90 % of $a_{y,max}$. The scenario space is specified in Table 2 by the min-max ranges of the five parameters from (3). The application space is slightly larger than the validation space so that it not only requires interpolation but also extrapolation capabilities. The values are chosen so that the manufactured universe reflects the real world properly.

We use a simple full factorial Design of Experiments (DoE) to generate the concrete scenarios. According to Table 2, 72 validation scenarios and 240 application scenarios are selected to legitimize the model-based process. In the future, it is possible to increase the amount of application scenarios and to select more sophisticated DoE techniques to gain efficiency. The crosses in Figure 4 show a two-dimensional cross-section along the velocity and the lateral acceleration dimension with a grid of nine validation and 30 application scenarios. We use this type of visualization to reflect the focus of the R-79 on the two parameters. All N LKFT scenarios will be aggregated in a data matrix

$$\mathbf{X} = \begin{bmatrix} 1 & v_1 & a_1 & v_{w,1} & s_{l,1} & l_{t,1} \\ 1 & v_2 & a_2 & v_{w,2} & s_{l,2} & l_{t,2} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & v_N & a_N & v_{w,N} & s_{l,N} & l_{t,N} \end{bmatrix} \in \mathbb{R}^{N \times (N_x + 1)} \quad (4)$$

for each domain: $\mathbf{X}^v \in \mathbb{R}^{N_x \times N^v}$ for validation and $\mathbf{X}^a \in \mathbb{R}^{N_x \times N^a}$ for the application.

5.3. Uncertainty Quantification and Propagation

A deterministic simulation performs a point prediction, since it assumes that all parameters are precisely known. A non-deterministic simulation, however, considers uncertain parameters, quantifies them, propagates them through the simulation model and aggregates the results. The uncertainties are summarized in Table 2 for the five scenario parameters. The values were systematically derived depending on how accurately it is possible to measure them during real validation experiments. We assume tight normal distributions for the velocity and lateral acceleration, since both can be measured accurately with Inertial Measurement Units (IMUs), and for the tank load, since the vehicle can be weighed in advance and the fuel level is available. We assume a coarser normal distribution for the wind – based on

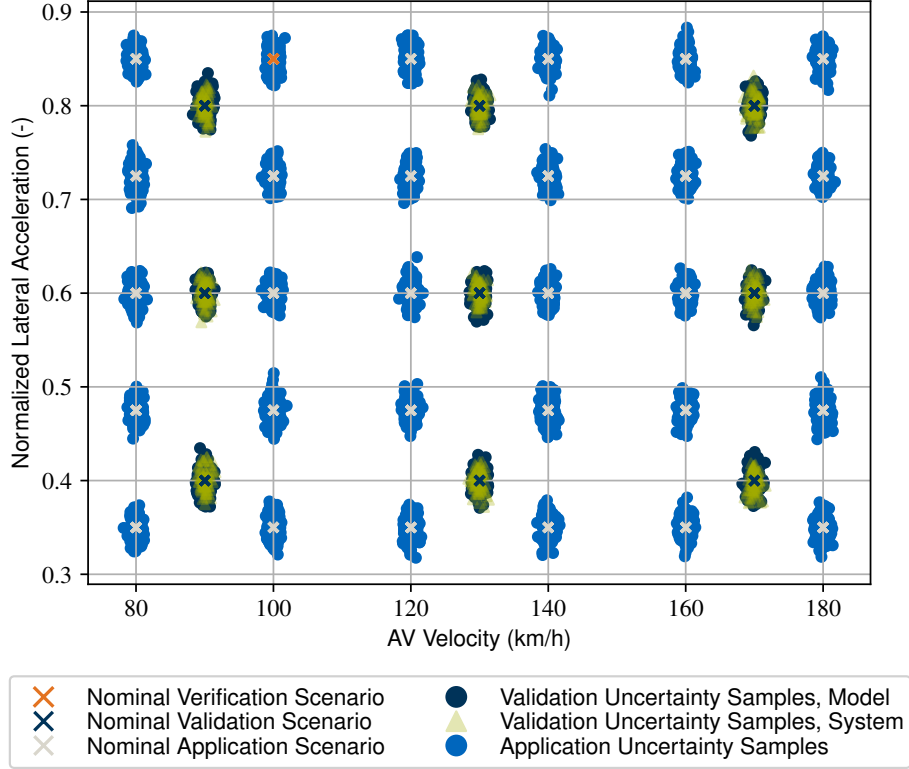


Figure 4: Verification, validation and application scenarios

online measuring stations – and an epistemic uncertainty for the road gradient, since it has to be extracted from map data of highways. This lack of knowledge from publicly accessible maps could only be compensated by extensive measurements. We assume further scenario parameters and the internal vehicle parameters as deterministic for this simplified proof of concept, since they are often precisely known and constant. This assumption is reflected in the model-form uncertainty and can be substantiated by a sensitivity analysis in the future. In addition, we assume the uncertainties from Table 2 to be equal around each validation and application scenario.

We use $N_r^v = 10$ repetitions of each validation scenario for the manufactured universe to imitate physical experiments and to capture their natural variability. The value is based on the findings of Viehof [41, p. 74], who recommends 10 to 15 repetitions based on the t-distribution for a detailed analysis and later at least three repetitions for practical experiments. These tests are the baseline for both the deterministic and the non-deterministic re-simulation to enable a fair comparison later. In the deterministic case, we re-simulate the mean value of the scenario repetitions (see averaged input case in Section 2.2.2). In the non-deterministic case, we apply the nested two-loop sampling approach from Section 2.3 to propagate the uncertainties. We use a full factorial design with three steps of the epistemic parameter in the outer loop and Monte Carlo sampling with ten random samples for all aleatory parameters in the inner loop. This is equivalent to a resolution in 10 % steps and is sufficient for the proof of concept in this paper. In total, we perform 2160 runs with the simulation model in the validation domain and 7200 runs in the application domain. The uncertainties are also illustrated in Figure 4 by projecting all samples of all parameters onto the velocity and lateral acceleration dimension. They form local point clouds within the large scenario space.

5.4. LKFT Assessment

Each simulation run results in one distance to line signal. It shows a typical behavior as shown in Figure 5, where the vehicle is pushed hardest to the edge at the entrance of the curve (cross at minimum distance to line), and then compensates a bit – unless it crosses the line at high lateral accelerations. We extract the minimum distance to the line

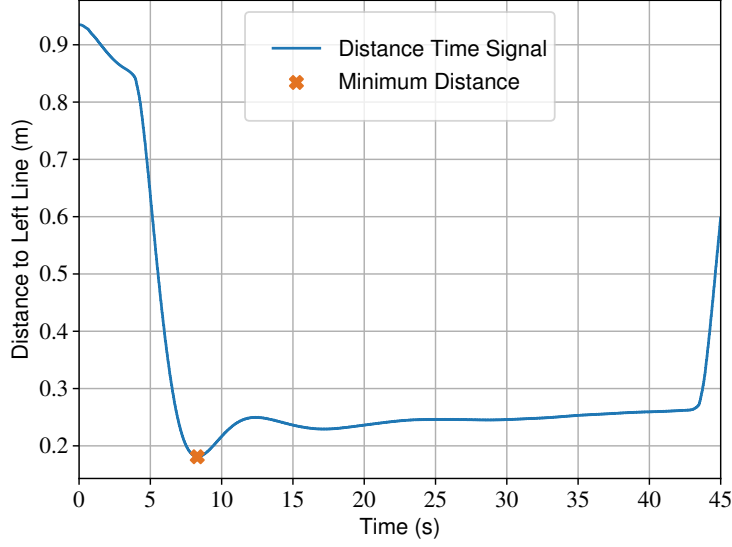


Figure 5: Characteristic lane keeping behavior by means of the distance to line time signal and its minimum value as KPI at the exemplary application scenario $v = 100$ km/h and $a = 0.35 \cdot a_{y,smax} = 0.875$ m/s².

as a KPI because it is the most safety critical. If not even the minimum distance falls below zero (crosses the line), the whole time signal will not fall below it. The behavior over the whole scenario space was already shown in Figure 2.

In the deterministic case, we average the repetition results so that we can use a scalar validation metric. In the non-deterministic case, we aggregate the repetitions to an ECDF and the nested uncertainty propagation to a p-box for the use of a non-deterministic metric.

5.5. Validation Metric

In the deterministic case, we use the absolute deviation between the minimum distance to line from the input-averaged re-simulation and the output-averaged minimum distance to line from the manufactured universe

$$e^v = g_m(\langle \mathbf{x}^v \rangle, \theta_m) - \langle g_s(\mathbf{x}^v, \theta_s) \rangle \quad (5)$$

as validation metric with $\langle \rangle$ as mean operator. In the non-deterministic case, we quantify the difference between the ECDF of the manufactured universe $F(y_s^v)$ and the p-box from the simulation model $B(y_m^v)$. There are different metrics that give different weight to different characteristics of a distribution. We use an Area Validation Metric (AVM) based on Voyles and Roy [59] because it includes the entire shape of the distributions in the calculation. The principle is illustrated in Figure 6a for one validation scenario. Since the manufactured universe shows worse performance, it only exists a left-hand area

$$e_l^v = \int_{B(y_m^v) \geq F(y_s^v)} |B(y_m^v) - F(y_s^v)| dy \quad (6)$$

where the p-box is larger than the ECDF. The area can be determined mathematically by integration of the step functions. Since the right-hand area is zero, we get a one-sided adaption of the area validation metric (without safety factor from [59]) in form of the interval of e^v (no function)

$$I(e^v) := \{e^v \mid \underline{e}^v \leq e^v \leq \bar{e}^v\} := [\underline{e}^v, \bar{e}^v] = [e_l^v, 0]. \quad (7)$$

5.6. Error Learning and Inference

It is essential to take into account the modeling errors and uncertainties for the final decision making in the application domain. The validation metric quantifies the model-form uncertainty for each validation scenario. Ultimately,

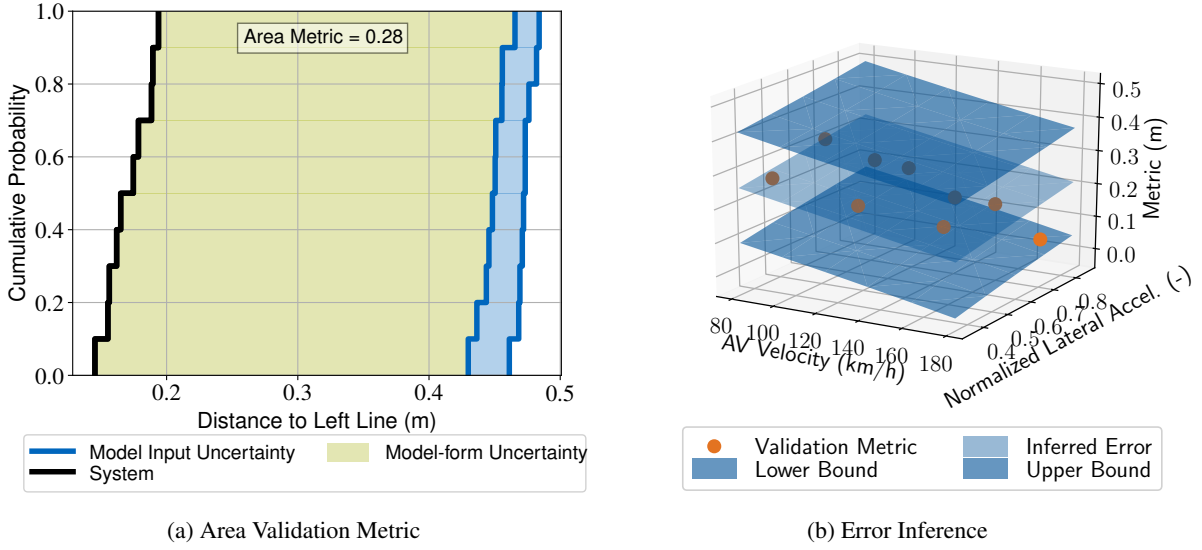


Figure 6: Validation errors. (a) Non-deterministic area metric for the exemplary validation scenario $v = 90$ km/h and $a = 0.4 \cdot a_{y,max} = 1$ m/s². (b) Linear regression inference of the deterministic validation error $\hat{e}^{va}$ with prediction intervals across the entire application scenario space.

however, we need to predict it for each application scenario. Therefore, we learn an error model with the metric results from the validation domain, so that we can use it afterwards to infer the errors in the application domain.

For this first proof of concept, we apply a multiple linear regression technique based on [50, p. 657] to model the validation error

$$\hat{e}^v = \mathbf{w} \cdot \mathbf{x}, \quad \mathbf{w} = \begin{bmatrix} w_0 & \cdots & w_{N_x} \end{bmatrix} \in \mathbb{R}^{N_x+1} \quad (8)$$

of the distance to line with the scenario parameters $\mathbf{x}$ as regressors. The regression weights $\mathbf{w}$ are determined so that the difference between the estimated and the actual validation errors is minimized

$$\arg \min_{\mathbf{w}} \sum_{i=1}^{N^v} (\hat{e}_i^{vv} - e_i^{vv})^2 \quad (9)$$

across the whole validation data set from Table 2 with N^v distinct scenarios. We emphasize estimates of the validation error in the validation domain with $\hat{e}^{vv}$ and estimates of the validation error in the application domain with $\hat{e}^{va}$. To get to the latter, we fit one linear model to infer the deterministic validation error $\hat{e}^{va}$, whereas we fit two linear models to infer both the left and right (zero here) interval boundaries of the non-deterministic model-form uncertainty $I(\hat{e}^{va}) = [\underline{e}^{va}, \bar{e}^{va}]$.

In addition, we calculate prediction intervals (PIs) based on [50, p. 657] to incorporate the uncertainty of the fitted model and the one associated with future observations. Since the prediction variable represents a random variable and we need an interval estimate for its mean value, we apply a non-simultaneous Bonferroni-type prediction interval via the function [60, p. 115]

$$g_p(\mathbf{x}^a) = t_{N^v - (N_x + 1)}^{\alpha/2} \cdot s \cdot \sqrt{1 + \mathbf{x}^{aT} (\mathbf{X}^{vT} \mathbf{X}^v)^{-1} \mathbf{x}^a} \quad (10)$$

with a confidence of $\alpha = 95\%$ for the t-distribution and with the sample variance

$$s^2 = \frac{1}{N^v - (N_x + 1)} \sum_{i=1}^{N^v} (e_i^{vv} - \hat{e}_i^{vv})^2. \quad (11)$$

In the deterministic case, the prediction interval $\pm g_p(\mathbf{x}^a)$ shifts the signed estimate of the error in both directions:

$$I(\hat{e}^{va}) = [\underline{e}^{va}, \bar{e}^{va}] = [\hat{e}^{va} - g_p(\mathbf{x}^a), \hat{e}^{va} + g_p(\mathbf{x}^a)]. \quad (12)$$

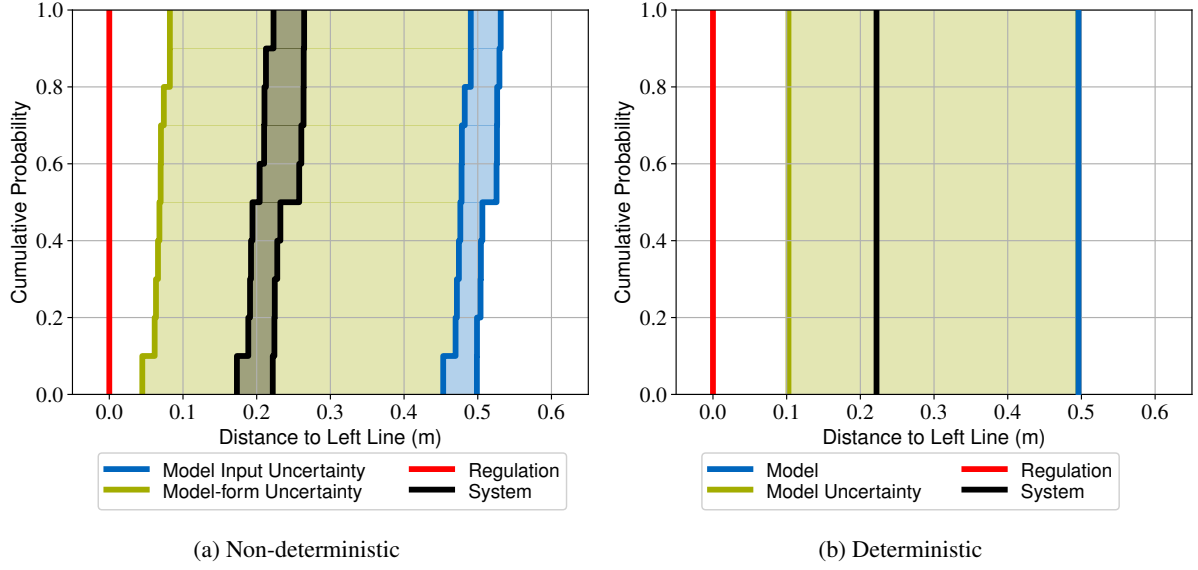


Figure 7: Uncertainty aggregation and decision making at the exemplary application scenario $v = 100 \text{ km/h}$ and $a = 0.35 \cdot a_{y,smax} = 0.875 \text{ m/s}^2$ based on the (a) deterministic point prediction and (b) non-deterministic p-box prediction of the nominal simulation model.

In the non-deterministic case, we add the respective function value to the left and right (positive) areas

$$I(\hat{e}^{va}) = [\underline{e}^{va}, \bar{e}^{va}] = [\hat{e}_l^{va} + g_{p,l}(\mathbf{x}^a), \hat{e}_r^{va} + g_{p,r}(\mathbf{x}^a)], \quad (13)$$

since the resulting outer bounds enclose the regression estimates and the inner bounds. Figure 6b illustrates the regression surface of the deterministic metric including prediction intervals. The light blue plane of the linear model follows the trend of the orange metric results used for training.

5.7. Uncertainty Expansion

After quantifying the individual errors and uncertainties, they must be aggregated in the application domain. In the non-deterministic case, we get for each application scenario one input uncertainty in the form of a p-box $B(y_m^a) = [F(y_m^a), \bar{F}(y_m^a)]$, one inferred model-form uncertainty in the form of an interval $I(\hat{e}^{va}) = [\underline{e}^{va}, \bar{e}^{va}]$ and one numerical uncertainty (zero here) in the form of an interval $I(\hat{e}^{na}) = [\underline{e}^{na}, \bar{e}^{na}]$. The combination of the uncertainties can be seen in Figure 7a for one application scenario. The two lower interval boundaries shift the input uncertainty p-box to the left, the two upper interval boundaries to the right. The resulting p-box reflects an estimate of the actual system response

$$B(\hat{y}_s^a) = \{F(\hat{y}_s^a) \mid F(y_m^a + (\underline{e}^{va} + \underline{e}^{na})) \leq F(\hat{y}_s^a) \leq \bar{F}(y_m^a - (\bar{e}^{va} + \bar{e}^{na}))\}. \quad (14)$$

In the example, the model-form uncertainty has the largest influence on the total uncertainty, since we injected a systematic error into the manufactured universe.

In the deterministic case, we use the inferred error bounds $I(\hat{e}^{va}) = [\underline{e}^{va}, \bar{e}^{va}]$ to adapt the actual model prediction y_m^a to a non-deterministic interval estimate of the system response

$$I(\hat{y}_s^a) = \begin{cases} [y_m^a - \bar{e}^{va}, y_m^a] & \underline{e}^{va}, \bar{e}^{va} \geq 0 \\ [y_m^a - \bar{e}^{va}, y_m^a - \underline{e}^{va}] & \underline{e}^{va} \leq 0, \bar{e}^{va} \geq 0 \\ [y_m^a, y_m^a - \underline{e}^{va}] & \underline{e}^{va}, \bar{e}^{va} \leq 0 \end{cases} \quad (15)$$

depending on different cases. If both error bounds were either greater or less than zero, the middle case would lead to a bias correction of the nominal model in one direction. This would be equivalent to putting more trust into the

data-driven error model than into the physical simulation model itself and would again be associated with uncertainties. To avoid this, we add two more cases, which can be interpreted as one-sided uncertainty expansion, starting from the simulation model as a stable baseline. The principle is also visualized in Fig. 7b. This mixed approach neither neglects the errors as usual, nor does it perform a risky bias correction, but it adds tight bounds for a fair comparison.

5.8. Decision Making

Fig 7a and Fig. 7b also include the regulation threshold as a vertical line, which states that the distance to line must be greater than zero (the vehicle edge does not cross the line). It transforms the real values into boolean decisions, because, in the end, the type approval requirements can only be passed or failed. Both in the deterministic and in the non-deterministic case, the entire green area exceeds the threshold and passes the requirements. This even corresponds to the most conservative non-deterministic estimate. A reduction of the confidence level of the CDF in the future would lower the requirements so that not all steps have to pass. Currently, checking all steps refers to a confidence of 90 % due to the 10 aleatory samples from Table 2. In total, the AV passes 123 and fails 117 of the 240 application scenarios in the deterministic case, whereas it passes 97 and fails 143 in the non-deterministic case. The values can be taken later as part of Table 3. The latter ones are, as expected, somewhat more conservative due to the consideration of the input uncertainties. Ultimately, the amount of failed decisions would lead in summary to a failure of the entire homologation.

6. Validation of the Application Results

The last section presented the VV&UQ methodology using an example where the type approval of a lane keeping assistant was carried out based on simulation. It concluded that the AV fails the type approval. However, what does this actually mean regarding the quality of the methodology itself? To answer this question, the Method of Manufactured Universes comes into play, where the actual ground truth values are available in all domains. The first subsection introduces the principle, how those values can be used to analyze the meaningfulness of the VV&UQ results. The second sub-section evaluates the tolerance approach as a baseline from the current state of the art. The two following subsections are dedicated to the validation and comparison of the deterministic and the non-deterministic manifestation of the framework from the last section.

6.1. Evaluation Methodology

The analyses in this section build up on the Method of Manufactured Universes. Its big advantage is that the actual Ground Truth (GT) values are available for all validation scenarios and in particular for all application scenarios. This makes it possible to compare the VV&UQ results with the true values and thus to evaluate the selected VV&UQ methodology itself. We simulate the manufactured universe in the application domain under the same conditions as the nominal simulation model. Even if additional real experiments were conducted to validate the methodology, this would never be possible. There will always be measurement errors or poor input uncertainties that falsify the ground truth values. These are factors that can be considered in the future to analyze the robustness of the approach against unexpected uncertainties. In the application domain, however, we require two p-boxes for the model and the manufactured universe under the same desired application conditions for a fair comparison.

The comparison can take place at multiple stages along the processing pipeline of the framework:

1. Error inference stage
2. Assessment stage
3. Error integration stage
4. Type approval stage.

We focus on the final two stages because they depend on the previous ones and are ultimately decisive. Starting backwards, the final binary decisions from the type approval can be compared. For this purpose a binary classifier is used in pattern recognition and machine learning. On the one hand, it distinguishes positive from negative decisions based on the absolute simulation results. On the other hand, it distinguishes true from false decisions based on the relation of simulation and experimental results. In our example, a True Positive (TP) stands for a correctly failed type approval, whereas a True Negative (TN) stands for a correctly passed type approval. On the contrary, a False Positive

(FP) stands for an incorrectly failed type approval and a False Negative (FN) for an incorrectly passed type approval. FPs are Type I errors that convict the innocent AV. FNs are Type II errors that acquit an unsafe AV. In addition, the precision P is based on the former and the recall R on the latter:

$$P = \frac{\sum TP}{\sum TP + \sum FP} \quad (16)$$

$$R = \frac{\sum TP}{\sum TP + \sum FN} . \quad (17)$$

Recall is a very important measure from the safety perspective to quantify how many of the actual approval fails are detected by the simulation model. Precision is also important to identify the approval fails with as few Type I errors as possible.

In the error integration stage, it is additionally possible to check whether the actual GT values fall within the bounds from the VV&UQ method. This looks as follows in the deterministic and the non-deterministic case:

$$y_s^a \in I(\hat{y}_s^a) \quad (18)$$

$$F(y_s^a) \in F(\hat{y}_s^a) \quad (19)$$

It gives additional insights and confidence in the error model, since it is possible to correctly classify the binary decisions depending on the distance to the threshold, despite the GT not falling within the bounds.

6.2. Tolerance Approach

It is important to emphasize that the pure tolerance approach from Section 2.2.2, as it is currently used in the automotive sector, is not directly transferable to controlled AVs. First of all, there are absolute and relative tolerances. The absolute ones struggle in the sense that they do not relate the validation errors to the scenario results. An exemplary assumed model error of 30 cm has much stronger effects when the virtual vehicle drives 15 cm beside the line than when it drives in the middle of the road. The final binary results for the pure tolerance approach (without error pipeline) for the absolute tolerance of 30 cm can be seen in Figure 8. It proves cases where the model is assumed to be valid according to the tolerances in the validation domain, but false classifications (FP and FN) between the nominal model and the GT occur at neighboring locations in the application domain. They are concerning from a safety perspective. There is also an inverse case with an invalid model and true classifications in the neighborhood that leaves potential. Classical relative tolerances suffer from similar problems. Suppose one allows a 10 % deviation of the jerk. Then larger jerk values benefit, although these are safety critical.

It would be possible to extend the tolerances to the safety perspective by dividing the validation errors by the distance from the threshold. Regardless of this, we recommend using the tolerances in the validation domain as a tool for the model developer, but not standalone. It should be combined with the vertical error pipeline in the framework in Figure 3 to consider model-form and extrapolation uncertainties. The use of tolerances alone separates the validation and application domain, neglects uncertainties and is risky.

6.3. Deterministic and Non-Deterministic Manifestation

This subsection continues with the analysis of the overall framework including the error and uncertainty pipeline as presented in Section 5. Table 3 summarizes the results from the binary classifier. The first column refers to the deterministic manifestation of the framework and the second column to the non-deterministic manifestation. In the first row, the classifier is used to compare the nominal simulation model with the GT from the manufactured universe, and in the second row, the results of the VV&UQ methodology are compared with the GT. The injected systematic error results in many safety critical FNs and a low recall rate with a perfect precision of 100 % in the Table 3a and 3b.

Checking whether the GT falls within the bounds of the VV&UQ methodology according to Equation (18) and (19) yields 238 bounded cases for the deterministic manifestation and even 240 of 240 bounded cases for the non-deterministic manifestation. This shows a working conservative error modeling using linear regression and prediction intervals. In both Table 3c and 3d, the VV&UQ methodology successfully shifts all the FNs into the TP cell. This heavily increases the recall from almost 0 to 100 %. From a safety perspective this is optimal since no unsafe AV is acquitted anymore. In return, a few TNs are moved into the FP cell, which convicts the innocent AV. Over the entire

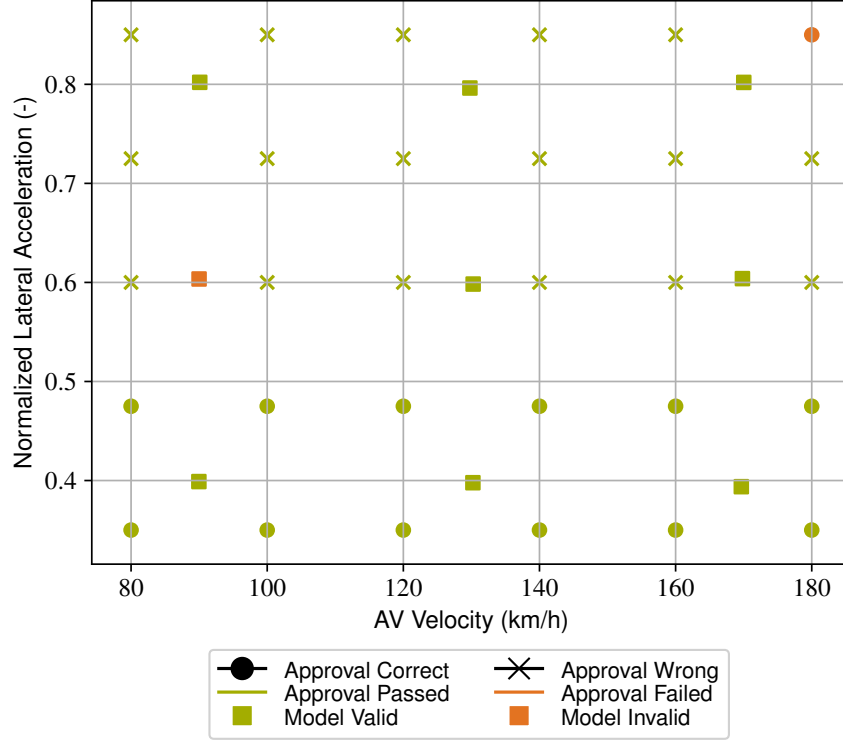


Figure 8: Decisions across the scenario space. The rectangles refer to general model validity based on the tolerance approach, the circles and crosses to the validity of the application decisions using the binary classifier. The decision of the nominal simulation model is coded in the color channel, the comparison with the ground truth values by the two symbols cross and circle. Thus, an orange circle symbolizes a TP, an orange cross an FP, a green cross an FN and a green circle a TN.

scenario space, this results in a precision of 77 % resp. 86 %. However, since the vehicle will not pass the homologation anyway due to the many TPs, the additional FPs do not matter much in this case. Generally, it is impossible to guarantee model validity with only a restricted amount of validation experiments [41]. Nevertheless, further cases and alternative metamodeling techniques should be considered in the future to address the trade-off between Type I and II errors.

6.4. Comparison and Discussion

Comparing the deterministic and the non-deterministic manifestation of the framework regarding the binary classifications shows similar results. The non-deterministic approach has a slight advantage due to a higher precision at the same recall rate. In principle, the classifier results allow both approaches and do not show a clear selection. Thus, further factors may be included in the process to select the configuration of the framework for the specific use case of this paper. The non-deterministic approach requires a higher effort to quantify the uncertainties and to perform the simulations. In return, it considers different sources of uncertainty including the scenario parameters. This results in a higher confidence in the statement of each application scenario and a larger coverage of the scenario space due to the scatter around each nominal scenario point. The accuracy of the statement can even be adjusted by the number of aleatory samples. In addition, the illustrative example with the comparatively large systematic error makes the strengths of the non-deterministic approach in the quantification of input uncertainties less obvious. Therefore, if the resources are available for the non-deterministic approach, its selection is recommended. Otherwise, the deterministic approach represents a good compromise between effort and risk.

Generally, we recommend to use all (5 here) uncertain parameters also as scenario parameters and to include them into the scenario design. It fits the nature of model validation to quantify the input uncertainties as accurately as possible around a nominal scenario condition. If an uncertain parameter is not included in the scenario design, its global uncertainty must be taken into account (space columns instead of the uncertainty column in Table 2). This leads

Table 3: Binary classifier

(a) Deterministic results – Nominal model				
Universe				
		Fails	Passes	
Model	Fails	2 TPs	0 FPs	$P = 100 \%$
	Passes	88 FNs	150 TNs	
$R \approx 2 \%$				

(b) Non-deterministic results – Nominal model				
Universe				
		Fails	Passes	
Model	Fails	5 TPs	0 FPs	$P = 100 \%$
	Passes	92 FNs	143 TNs	
$R \approx 5 \%$				

(c) Deterministic results – VV&UQ method				
Universe				
		Fails	Passes	
VVUQ	Fails	90 TPs	27 FPs	$P \approx 77 \%$
	Passes	0 FNs	123 TNs	
$R = 100 \%$				

(d) Non-deterministic results – VV&UQ method				
Universe				
		Fails	Passes	
VVUQ	Fails	123 TPs	20 FPs	$P \approx 86 \%$
	Passes	0 FNs	97 TNs	
$R = 100 \%$				

to wider p-boxes in case of larger input uncertainties and thus to an under-approximation of the model error. This in turn is too little to perform a sufficient shift of the model responses during the uncertainty expansion and would finally lead to some false classifications.

7. Conclusion

For credible safeguarding of automated vehicles based on computer simulation, model validation activities are essential to assess the errors and uncertainties inherent in every model. Therefore, we presented a modular framework based on the type approval of a lane keeping function. Due to the modular design, the framework can be used in different manifestations depending on the requirements. On the one hand, we presented a non-deterministic manifestation considering several types of uncertainties. On the other hand, we presented a simplified deterministic manifestation that still considers an overall model error. We analyzed both manifestations in regard to ground truth values using a binary classifier.

Both approaches show excellent results in the identification of systematic errors. They corrected all safety-critical cases where the nominal simulation model suggests a suitable lane keeping behavior although the actual ground truth illegally crosses the line. The non-deterministic approach shows slightly better results, but is also more complex. In the future, we plan to analyze different scenario designs, validation metrics and error learning techniques to further improve the results. Nevertheless, it is important to mention that there is no absolute model validity since it is impossible to guarantee validity across the entire scenario space with only a restricted amount of validation experiments. In addition, we can apply the framework in the next step to a real vehicle, after we have validated the methodology in this paper based on the method of manufactured universes.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Contribution

Stefan Riedmaier initiated and wrote this paper. He was involved in all stages of development and primarily developed the concept and content of this work. Jakob Schneider wrote his master thesis about uncertainties in safeguarding an ACC. Afterwards, he continued his research by supporting Stefan Riedmaier in implementing the framework, in applying it to the LKAS type approval and in enhancing the content in frequent workshops. Benedikt Danquah contributed to the structure of the paper and improved the content thanks to a close cooperation and many valuable discussions on VV&UQ methods. Bernhard Schick and Frank Diermeyer contributed to the conception of the research project and revised the paper critically for important intellectual content. Frank Diermeyer gave final approval of the version to be published and agrees to all aspects of the work. As a guarantor, he accepts responsibility for the overall integrity of the paper.

References

- [1] SAE J3016, Taxonomy and Definitions for Terms Related to Driving Automation Systems for On-Road Motor Vehicles, 2018.
- [2] H. Beglerovic, M. Stolz, M. Horn, Testing of autonomous vehicles using surrogate models and stochastic optimization, in: 2017 IEEE 20th International Conference on Intelligent Transportation Systems (ITSC), IEEE, 2017, pp. 1–6.
- [3] United Nations Economic Commission for Europe (UNECE), Proposal for a new un regulation on uniform provisions concerning the approval of vehicles with regards to automated lane keeping system (ece/trans/wp.29/2020/81), 2020.
- [4] N. Kalra, S. M. Paddock, Driving to safety: How many miles of driving would it take to demonstrate autonomous vehicle reliability?, *Transportation Research Part A: Policy and Practice* 94 (2016) 182–193. URL: <http://www.sciencedirect.com/science/article/pii/S0965856416302129>.
- [5] W. Wachenfeld, H. Winner, The Release of Autonomous Vehicles, in: M. Maurer, J. C. Gerdes, B. Lenz, H. Winner (Eds.), *Autonomous Driving: Technical, Legal and Social Aspects*, Springer Berlin Heidelberg, Berlin, Heidelberg, 2016, pp. 425–449. URL: https://doi.org/10.1007/978-3-662-48847-8_21.
- [6] German Aerospace Center, PEGASUS-project, 2019. URL: <https://www.pegasusprojekt.de/en/home>.
- [7] A. Leitner, A. Akkermann, B. Å. Hjøllø, B. Wirtz, D. Nickovic, E. Möhlmann, H. Holzer, J. van der Voet, J. Niehaus, M. Sarrazin, M. Zofka, M. Rooker, M. Kubisch, M. Paulweber, M. Siegel, M. Rautila, N. Marko, P. Tummeltshammer, P. Rosenberger, R. Rott, S. Muckenhuber, S. Kalisvaart, T. d. Graaff, T. D'Hondt, T. Fleck, Z. Slavik, Enable-s3: testing & validation of highly automated systems: Summary of results, 2019.
- [8] E. F. Z. Santana, G. Covas, F. Duarte, P. Santi, C. Ratti, F. Kon, Transitioning to a driverless city: Evaluating a hybrid system for autonomous and non-autonomous vehicles, *Simulation Modelling Practice and Theory* 107 (2021) 102210.
- [9] UL, UL 4600: Standard for Safety for the Evaluation of Autonomous Products, 2019.
- [10] B. Schürmann, D. Heß, J. Eilbrecht, O. Stursberg, F. Koster, M. Althoff, Ensuring drivability of planned motions using formal methods, in: 2017 IEEE 20th International Conference on Intelligent Transportation Systems (ITSC), IEEE, 2017, pp. 1–8.
- [11] S. Kitajima, K. Shimono, J. Tajima, J. Antona-Makoshi, N. Uchida, Multi-agent traffic simulations to estimate the impact of automated technologies on safety, *Traffic injury prevention* 20 (2019) 58–64.
- [12] S. Riedmaier, T. Ponn, D. Ludwig, B. Schick, F. Diermeyer, Survey on scenario-based safety assessment of automated vehicles, *IEEE Access* 8 (2020) 87456–87477.
- [13] S. Riedmaier, B. Danquah, B. Schick, F. Diermeyer, Unified framework and survey for model verification, validation and uncertainty quantification, *Archives of Computational Methods in Engineering* (2020).
- [14] M. G. Faes, M. A. Valdebenito, Fully decoupled reliability-based design optimization of structural systems subject to uncertain loads, *Computer Methods in Applied Mechanics and Engineering* 371 (2020) 1–17.
- [15] M. Eek, H. Gavel, J. Ölvander, Definition and implementation of a method for uncertainty aggregation in component-based system simulation models, *Journal of Verification, Validation and Uncertainty Quantification* 2 (2017) pp. 011001–1–011001–12.
- [16] H. F. Stripling, M. L. Adams, R. G. McClarren, B. K. Mallick, The method of manufactured universes for validating uncertainty quantification methods, *Reliability Engineering & System Safety* 96 (2011) 1242–1256.
- [17] R. G. Hills, Roll-up of validation results to a target application, 2013.
- [18] American Society of Mechanical Engineers, Standard for verification and validation in computational fluid dynamics and heat transfer: An American national standard, volume 20-2009 of *ASME V&V*, reaffirmed 2016 ed., The American Society of Mechanical Engineers, New York, NY, 2009.
- [19] J. Mullins, Y. Ling, S. Mahadevan, L. Sun, A. Strachan, Separation of aleatory and epistemic uncertainty in probabilistic model validation, *Reliability Engineering & System Safety* 147 (2016) 49–59.

- [20] S. Sankararaman, S. Mahadevan, Integration of model verification, validation, and calibration for uncertainty quantification in engineering systems, *Reliability Engineering & System Safety* 138 (2015) 194–209.
- [21] S. Atamturktur, F. M. Hemez, J. A. Laman, Uncertainty quantification in model verification and validation as applied to large scale historic masonry monuments, *Engineering Structures* 43 (2012) 221–234.
- [22] H. Abbas, M. O’Kelly, A. Rodionova, R. Mangharam, Safe at any speed: A simulation-based test harness for autonomous vehicles, in: *Seventh Workshop on Design, Modeling and Evaluation of Cyber Physical Systems (CyPhy’17)*, 2017, pp. 94–106.
- [23] T. Hanke, A. Schaermann, M. Geiger, K. Weiler, N. Hirsenkorn, A. Rauch, S.-A. Schneider, E. Biebl, Generation and validation of virtual point cloud data for automated driving systems, in: *2017 IEEE 20th International Conference on Intelligent Transportation Systems (ITSC)*, IEEE, 2017, pp. 1–6.
- [24] A. Schaermann, A. Rauch, N. Hirsenkorn, T. Hanke, R. Rasshofer, E. Biebl, Validation of vehicle environment sensor models, in: *2017 IEEE Intelligent Vehicles Symposium (IV)*, IEEE, 2017, pp. 405–411.
- [25] A. Gaidon, Q. Wang, Y. Cabon, E. Vig, Virtualworlds as proxy for multi-object tracking analysis, in: *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, IEEE, 2016, pp. 4340–4349.
- [26] M. Nentwig, M. Miegler, M. Stamminger, Concerning the applicability of computer graphics for the evaluation of image processing algorithms, in: *2012 IEEE International Conference on Vehicular Electronics and Safety (ICVES 2012)*, IEEE, 2012, pp. 205–210.
- [27] E. L. Zec, N. Mohammadiha, A. Schliep, Statistical sensor modelling for autonomous driving using autoregressive input-output hmms, in: *2018 IEEE 21th International Conference on Intelligent Transportation Systems (ITSC)*, IEEE, 2018, pp. 1331–1336.
- [28] International Organization for Standardization, Passenger cars — validation of vehicle dynamic simulation — sine with dwell stability control testing, 2016.
- [29] International Organization for Standardization, Passenger cars — vehicle dynamic simulation and validation — steady-state circular driving behaviour, 2016.
- [30] E. Kutluay, H. Winner, Validation of vehicle dynamics simulation models – a review, *Vehicle System Dynamics* 52 (2014) 186–200.
- [31] K. Groh, S. Wagner, T. Kuehbeck, A. Knoll, Simulation and its contribution to evaluate highly automated driving functions, in: *WCX SAE World Congress Experience*, SAE Technical Paper Series, SAE International 400 Commonwealth Drive, Warrendale, PA, United States, 2019, pp. 1–11.
- [32] D. Notz, M. Sigl, T. Kühbeck, S. Wagner, K. Groh, C. Schütz, D. Watzenig, Methods for improving the accuracy of the virtual assessment of autonomous driving, in: *2019 IEEE International Conference on Connected Vehicles and Expo (ICCVE) Proceedings*, 2019, pp. 1–6.
- [33] S. Wagner, K. Groh, T. Kuehbeck, A. Knoll, Towards cross-verification and use of simulation in the assessment of automated driving, in: *2019 IEEE Intelligent Vehicles Symposium (IV)*, IEEE, 2019, pp. 1589–1596.
- [34] B. Johnson, F. Havlak, H. Kress-Gazit, M. Campbell, Experimental evaluation and formal analysis of high-level tasks with dynamic obstacle anticipation on a full-sized autonomous vehicle, *Journal of Field Robotics* 34 (2017) 897–911.
- [35] S. Riedmaier, J. Nesensohn, C. Gutenkunst, T. Düser, B. Schick, H. Abdellatif, Validation of x-in-the-loop approaches for virtual homologation of automated driving functions, in: *11th Graz Symposium Virtual Vehicle (GSVF)*, 2018, pp. 1–12.
- [36] W. Daamen (Ed.), *Traffic simulation and data: Validation methods and applications*, [elektronische ressource] ed., Taylor and Francis and CRC Press, Hoboken and Boca Raton, Fla., 2015.
- [37] Y. Hollander, R. Liu, The principles of calibrating traffic microsimulation models, *Transportation* 35 (2008) 347–362.
- [38] L. Rao, L. Owen, Validation of high-fidelity traffic simulation models, *Transportation Research Record: Journal of the Transportation Research Board* 1710 (2000) 69–78.
- [39] T. Toledo, H. N. Koutsopoulos, Statistical validation of traffic simulation models, *Transportation Research Record: Journal of the Transportation Research Board* 1876 (2004) 142–150.
- [40] E. Kutluay, *Development and Demonstration of a Validation Methodology for Vehicle Lateral Dynamics Simulation Models*, Ph.D. thesis, Technische Universität Darmstadt, Darmstadt, 2012.
- [41] M. Viehof, *Objektive Qualitätsbewertung von Fahrdynamiksimulationen durch statistische Validierung*, Ph.D. thesis, Technische Universität Darmstadt, Darmstadt, 2018.
- [42] S. Rhode, Non-stationary gaussian process regression applied in validation of vehicle dynamics models, *Engineering Applications of Artificial Intelligence* 93 (2020) 103716.
- [43] M. Hartung, D. Hess, R. Lattarulo, J. Oehlerking, J. Perez, A. Rausch, Report on conformance testing of application models, 2017.
- [44] E. Böde, M. Büker, Ulrich Eberle, M. Fränzle, S. Gerwinn, B. Kramer, Efficient splitting of test and simulation cases for the verification of highly automated driving functions, in: B. Gallina, A. Skavhaug, F. Bitsch (Eds.), *Computer Safety, Reliability, and Security*, Springer International Publishing, Cham, 2018, pp. 139–153.
- [45] S. Detering, L. Schnieder, E. Schnieder, Two-level validation and data acquisition for microscopic traffic simulation models, *International Journal on Advances in Systems and Measurements* 3 (2010).
- [46] T. Ponn, T. Kröger, F. Diermeyer, Performance analysis of camera-based object detection for automated vehicles, *Sensors (Basel, Switzerland)* 20 (2020).
- [47] M. Holder, P. Rosenberger, H. Winner, V. P. Makkapati, M. Maier, H. Schreiber, Z. Magosi, T. D’hondt, Z. Slavik, O. Bringmann, W. Rosenstiel, Measurements revealing challenges in radar sensor modeling for virtual validation of autonomous driving, in: *2018 IEEE 21th International Conference on Intelligent Transportation Systems (ITSC)*, IEEE, 2018, pp. 2616–2622.
- [48] P. Rosenberger, M. Holder, M. Zirulnik, H. Winner, Analysis of real world sensor behavior for rising fidelity of physically based lidar sensor models, in: *2018 IEEE Intelligent Vehicles Symposium (IV)*, IEEE, 2018, pp. 611–616.
- [49] United Nations Economic Commission for Europe (UNECE), Addendum 139 - regulation no. 140 — uniform provisions concerning the approval of passenger cars with regard to electronic stability control (esc) systems, 2017.
- [50] W. L. Oberkamp, C. J. Roy, *Verification and Validation in Scientific Computing*, Cambridge University Press, Cambridge, 2010.
- [51] M. Aramrattana, R. H. Patel, C. Englund, J. Härrj, J. Jansson, C. Bonnet, Evaluating model mismatch impacting cacc controllers in mixed traffic using a driving simulator, in: *2018 IEEE Intelligent Vehicles Symposium (IV)*, IEEE, 2018, pp. 1867–1872.
- [52] L. Zheng, T. Sayed, M. Essa, Y. Guo, Do simulated traffic conflicts predict crashes? an investigation using the extreme value approach, in:

- 2019 IEEE 22th International Conference on Intelligent Transportation Systems (ITSC), IEEE, 2019, p. 631–636.
- [53] D. C. Kammer, P. A. Blelloch, J. Sills, Test-based uncertainty quantification and propagation using hurty/craig-bampton substructure representations, in: *Proceedings of the IMAC-XXXVII*, 2019, pp. 107–129.
 - [54] M. N. Avramova, K. N. Ivanov, Verification, validation and uncertainty quantification in multi-physics modeling for nuclear reactor design and safety analysis, *Progress in Nuclear Energy* 52 (2010) 601–614.
 - [55] United Nations Economic Commission for Europe (UNECE), Addendum 78: Un regulation no. 79 — uniform provisions concerning the approval of vehicles with regard to steering equipment, 2018.
 - [56] G. Bagschik, T. Menzel, M. Maurer, Ontology based scene creation for the development of automated vehicles, in: *2018 IEEE Intelligent Vehicles Symposium (IV)*, 2018, pp. 1813–1820. doi:10.1109/IVS.2018.8500632.
 - [57] N. W. Whiting, C. J. Roy, E. P. Duque, S. Lawrence, Assessment of model validation and calibration approaches in the presence of uncertainty, in: *AIAA Scitech 2019 Forum*, American Institute of Aeronautics and Astronautics, 2019, pp. 1–16.
 - [58] M. Viehof, H. Winner, Research methodology for a new validation concept in vehicle dynamics, *Automotive and Engine Technology* (2018).
 - [59] I. T. Voyles, C. J. Roy, Evaluation of model validation techniques in the presence of aleatory and epistemic input uncertainties, in: *17th AIAA Non-Deterministic Approaches Conference*, American Institute of Aeronautics and Astronautics, 2015, pp. 1–16.
 - [60] R. G. Miller, *Simultaneous Statistical Inference*, Springer New York, New York, NY, 1981.